Definition of Convergence in Distribution

Review: Cumulative Distribution Function

- Let X be a random variable. Then its **cdf** is defined by

$$F_X(x) = \mathbb{P}(X \leq x).$$

  - F is a nondecreasing function.
  - $\lim_{x \to -\infty} F(x) = 0$, $\lim_{x \to \infty} F(x) = 1$.
  - F is right continuous, $\lim_{x \downarrow x_0} F(x) = F(x_0)$.

Convergence in Distribution

- Let $\{X_n\}$ be a sequence of random variables and X be a random variable. Let F_{X_n} and F_X be, respectively, the cdfs of X_n and X. Let $C(F_X)$ denote the set of all points where F_X is continuous. We say that X_n **converges in distribution** to X if

$$\lim_{n \to \infty} F_{X_n}(x) = F_X(x) \quad \text{for all } x \in C(F_X).$$

We denote this convergence by

$$X_n \xrightarrow{D} X.$$

A Motivation Example

- Motivation for only considering points of continuity of F_X is given by following simple example. Let X_n be the random variable with all its mass at $1/n$ and let X be a random variable with all its mass at 0. At the point of discontinuity of F_X, $\lim F_{X_n}(0) = 0 \neq 1 = F_X(0)$, but $X_n \xrightarrow{D} X$.

A Motivation Example

Convergence in Distribution

- Convergence in distribution does **NOT** mean convergence in probability: Let X be a continuous random variable with a pdf $f_X(x)$ which is symmetric about 0; i.e., $f_X(-x) = f_X(x)$. Thus, X and $-X$ have the same distributions. Define X_n as

$$X_n = \begin{cases} X & \text{if n is odd} \\ -X & \text{if n is even.} \end{cases}$$

Clearly, $F_{X_n}(x) = F_X(x)$ and $X_n \xrightarrow{D} X$. However, $X_n \xrightarrow{P} X$

Examples

- Let $\bar{X}_n$ have the cdf

$$F_n(\bar{x}) = \int_{-\infty}^{\bar{x}} \frac{1}{\sqrt{1/n}\sqrt{2\pi}} e^{-nw^2/2} dw.$$

If the change of variable $v = \sqrt{n}w$ is made, we have

$$F_n(\bar{x}) = \int_{-\infty}^{\sqrt{n}\bar{x}} \frac{1}{\sqrt{2\pi}} e^{-v^2/2} dv.$$

It is clear that

$$\lim_{n\to\infty} F_n(x) = \begin{cases} 0 & \bar{x} < 0 \\ \frac{1}{2} & \bar{x} = 0 \\ 1 & \bar{x} > 0 \end{cases}$$

Now the function

$$F(\bar{x}) = \begin{cases} 0 & \bar{x} < 0 \\ 1 & \bar{x} \geq 0 \end{cases}$$

is a cdf and $\lim_{n\to\infty} F_n(x) = F(\bar{x})$ at every point of continuity of $F(\bar{x})$. So, $\bar{X}_n$ converges in distribution to a random variable that has a degenerate distribution at $\bar{x} = 0$.

Examples

- Even if a sequence $X_1, X_2, X_3, \ldots$ converges in distribution to a random variable X, we cannot in general determine the distribution of X by taking the limit of the pmf of X_n. Let X_n have the pmf

$$p_n(x) = \begin{cases} 1 & x = 2 + n^{-1} \\ 0 & \text{elsewhere.} \end{cases}$$

  Clearly, $\lim_{n\to\infty} p_n(x) = 0$ for all values of x. This may suggest that X_n does not converge in distribution.

Examples

However, the cdf of X_n is

$$F_n(x) = \begin{cases} 0 & x < 2 + n^{-1} \\ 1 & x \geq 2 + n^{-1}, \end{cases}$$

and

$$\lim_{n \to \infty} F_n(x) = \begin{cases} 0 & x \leq 2 \\ 1 & x > 2. \end{cases}$$

$$F(x) = \begin{cases} 0 & x < 2 \\ 1 & x \geq 2. \end{cases}$$

So, $X_1, X_2, X_3, \ldots$ converges in distribution to a random variable with cdf $F(x)$.

Examples

- **Maximum of a Sample from a Uniform Distribution (cont.)**: $X_1, \ldots, X_n \sim Unif(0, \theta)$. Let $Y_n = \max\{X_1, \ldots, X_n\}$. Consider the random variable $Z_n = n(\theta - Y_n)$. Let $t \in (0, n\theta)$. The cdf of Z_n is

$$P(Z_n \leq t) = P(Y_n \geq \theta - (t/n)) = 1 - (1 - \frac{t/\theta}{n})^n$$

$$\rightarrow 1 - e^{-t/\theta}$$

So, $Z_n \overset{D}{\rightarrow} Z$ where Z is distributed $\exp(\theta)$.

Some Theorems

- **Theorem 4.3.1.**: If X_n converges to X **in probability**, then X_n converges to X **in distribution**.
- In general, the converse of the above theorem is not true.
- **Theorem 4.3.2.**: If X_n converges to the constant b **in distribution**, then X_n converges to b **in probability**.
- **Theorem 4.3.3.**: Suppose X_n converges to X in distribution and Y_n converges in probability to 0. Then $X_n + Y_n$ converges to X in distribution.

Some Theorems

- **Theorem 4.3.4.**: Suppose X_n converges to X in distribution and g is a continuous function on the support of X. Then $g(X_n)$ converges to $g(X)$ in distribution.
- **Theorem 4.3.5. (Slutsky's Theorem)**: Let X_n, X, A_n, B_n be random variables and let a and b be constants. If $X_n \xrightarrow{D} X$, $A_n \xrightarrow{P} a$, and $B_n \xrightarrow{P} b$, then

$$A_n + B_n X_n \xrightarrow{D} a + bX.$$